Mathematics: Statistics & Analytic Number Theory.

PROJECT PROPOSAL

On a New Fundamental Class of Discrete Probability Distributions.
With an Application to Analytic Number Theory.

Applicant TBA

Institution Department of Statistics and Operations Research at the University of Vienna

Date 23 July 2026

Contents

1. State of the Art and Preliminary Work **3**

1.1. State of Research . . . 3

1.2. Own Preliminary Work . . . 4

2. Objectives and Research Questions **6**

2.1. Overall Objective . . . 6

2.2. Specific Research Questions . . . 6

2.3. Expected Results . . . 6

3. Statement of the Expected Novelty and Scientific Significance **8**

4. Methodology **9**

5. Work Plan and Research Schedule **11**

5.1. Phase I: Foundations and Initial Analysis . . . 11

5.2. Phase II: Development of the Main Theoretical Results . . . 11

5.3. Phase III: Extensions, Generalisations, and Validation . . . 12

5.4. Phase IV: Consolidation, Dissemination, and Completion . . . 13

6. Project-relevant Scientific Qualifications of the Researchers Involved **14**

7. Ethical, Safety-Related and Regulatory Aspects **15**

8. Sex and Gender Dimensions of the Project **16**

A. Appendix 1: References **17**

B. Appendix 2: Host Research Institution and Justification of Requested Funds **20**

C. Appendix 3: Scientific CVs and Overview of Previous Research Achievements **21**

D. Appendix 4: Statement on Environmental Sustainability in the Research Process **25**

1. State of the Art and Preliminary Work

We introduce a fundamental class of new discrete hyperbolic secant probability distributions. The hyperbolic secant is the reciprocal of the hyperbolic cosine

$$\cosh(x) = \frac{e^{-x} + e^{x}}{2}, \quad \operatorname{sech} = \frac{2}{e^{-x} + e^{x}}$$

which appears a lot in everyday context: A chain suspended between two road bollards follows a hyperbolic cosine curve. These fundamental distributions have not yet been systematically described in the literature. We can determine their normalisation constants based on [Ramanujan and Berndt, 1991]. We will analyse them systematically from a statistical perspective, apply them in the study of infinite series, and establish promising connections with analytic number theory. We can apply these new fundamental probability distributions to computer science. In particular, we can continue working on multidimensional digital searching in tries which was pioneered by [Flajolet and Puech, 1986] and continued e. g. in [Kirschenhofer, Prodinger and Szpankowski, 1996] at TU Vienna.

1.1. State of Research

To the best of our knowledge, the idea to use the hyperbolic secant for probability distributions dates earliest back to the paper [Fisher, 1921]. There, on p. 8, Fisher introduced the continuous hyperbolic secant distribution (HSD) via a probability measure $\operatorname{sech}(z-\zeta)/\pi \, \mathrm{d}z$ whose symmetric density is centered around $\zeta \in \mathbb{R}$. Later, Talacko analysed the HSD as a special case of the family of Perks' distributions in [Talacko, 1956]. In [Harkness and Harkness, 1968], the HSD was generalized to the generalized hyperbolic secant distribution (GHS) by adding parameters $\alpha, \varrho > 0$ to the characteristic function, viz. $\varphi(t) = \operatorname{sech}^{\varrho}(\alpha t)$.

They are defined via the characteristic function $\varphi(t) = \operatorname{sech}^{\varrho}(\alpha t)$ with parameters $\alpha, \varrho > 0$. For $\varrho = 1$, we get the classical HSD. Harkness and Harkness calculate the density functions and moments for these distributions. They also show the infinitely divisibility of the GHS distributions. Fischer gives in [Fischer, 2014] numerous further generalisations of the continuous HSD. In addition to the generalizations mentioned above, he discusses the Meixner, or NEF-GHS, family, which arises from the HS distribution as a natural exponential family; the Beta-HS distribution, which is obtained through a beta transformation; and the SHS and SASHS families, which result from further transformations, such as the sinh–arcsinh transformation. Finally, he applies the generalized HS models to financial data, using them, for example, as alternatives to the heavy-tailed stable distributions. Jiu and Peng provide a further continuous generalization in [Jiu and Peng, 2025]. They do not alter the characteristic function as Harkness and Harkness did, but they introduce parameters direct to the probability density function, viz. $\operatorname{sech}^{\vartheta_1}(x)e^{\vartheta_2 x}$. Nowadays, the standardised version of the HSD is most common.

All papers and Fischer's book treat continuous versions of the SH distribution. Nevertheless, they serve as foundational references for the statistical analysis and possible generalizations of our discrete HS distributions.

1.2. Own Preliminary Work

We already determined some normalisation constants for the new discrete hyperbolic secant (DHS) distributions from Ramanujan's series in his second notebook, cf. chapter 17 in [Ramanujan and Berndt, 1991]. Using Gauß' constant $G = \varpi/\pi$ with the lemniscate constant $\varpi = \Gamma^2(1/4)/(2\sqrt{2\pi})$ and the lemniscate modulus $k = 1/\sqrt{2}$, we can represent the probability mass functions of these distributions in very basic terms which directly reveals their fine structure, e. g. we get the probability mass function

$$(\mathbb{P}(X = k))_{k\in\mathbb{Z}} = \left(\frac{\operatorname{sech}(\pi k)}{\sqrt{2}G}\right)_{k\in\mathbb{Z}}$$

for the first bilateral discrete hyperbolic secant (DHS) distribution.

We also could generalize the fundamental DHS distributions to a large family of discrete distributions of the form $\{C_r \operatorname{sech}(\sqrt{r}\pi k)\}_{r\in\mathbb{N}}$. We used a Ramanujan series and applied the lambda-star function λ^* which yields the required singular values of the occurring elliptic integrals. E. g. we get the probability mass fucntion

$$\left(\frac{2^{4/3}\pi^2}{3^{1/4}\Gamma^3(1/3)}\operatorname{sech}(\sqrt{3}\pi k)\right)_{k\in\mathbb{Z}}.$$

We also calculated moments up to the order eight for the first DHS distribution, again relying on the work of Ramanujan in chapter 17 of [Ramanujan and Berndt, 1991], e. g.

$$\mathbb{E}\left[X^2\right] = \mathrm{Var}(X) = \frac{G^2}{4}.$$

We have not yet determined any characteristic functions, but we calculated the $-2\pi i$-Fourier transform for the first bilateral DHS distribution. This is very comfortable, since the hyperbolic secant is a fix-point of this Fourier transform. Therefrom, we could use the Poisson summation formula to calculate values of hyperbolic series up to the order nine, e. g.

$$\sum_{k\in\mathbb{Z}} \operatorname{sech}^3(\pi k) = \frac{G^3 + G}{\sqrt{2}},$$

and provide statistical interpretations for some of Ramanujan's series. Finally, we could apply the the first bilateral DHS distribution to analytic number theory. We calculated the exact value of the contour integral over the critical line $\Re(s) = 1/2$ from Riemann's hypothesis for a product of Dirichlet's beta function, the gamma function and Riemann's zeta function, viz.

$$\int_{1/2-i\cdot\infty}^{1/2+i\cdot\infty} \pi^{-(s+2)} \cdot \beta(s+2) \cdot \Gamma(s+2) \cdot \zeta(s)\,\mathrm{d}s = i\left(\frac{\varpi}{\sqrt{2}}\mathbb{E}\left[X^2\right] - \frac{\pi}{8}\right) = i\left(\frac{\varpi}{\sqrt{2}}\frac{G^2}{4} - \frac{\pi}{8}\right).$$

For the existence of this integral, we used a result from [Bourgain, 2016], in particular $\zeta(1/2 + i \cdot t) \in O\left(|t|^{\frac{13}{84}+\delta}\right)$ $(|t| \to \infty)$ for any $\delta > 0$ and Stirling's formula for the gamma function results in the convergence of the integral due to negative exponential growth.

Overall, we want to systematically widen our techniques to the other DHS distributions and investigate their relationship with infinite series and analytic number theory.

2. Objectives and Research Questions

2.1. Overall Objective

The objective of this project is to conduct a comprehensive statistical analysis of six new discrete hyperbolic secant (DHS) distributions and to investigate their connections with infinite series and analytic number theory.

2.2. Specific Research Questions

1. What are the characteristic functions and further statistical metrics of the six new DHS distributions?
2. How can the DHS distributions be used to calculate values of hyperbolic series and provide a statistical interpretation of some of Ramanujan's series?
3. How can Fourier series of the hyperbolic secant be employed to establish and investigate connections between the DHS distributions and analytic number theory?

2.3. Expected Results

The project is expected to produce at least the following results:

Rigorous definitions and normalization formulas for six new discrete probability distributions: We define the first, second and third discrete uni- and bilateral hyperbolic secant distributions (DHS distributions). The normalization constants can be derived from Ramanujan's second notebook, cf. chapter 17 of [Ramanujan and Berndt, 1991] by using some elliptic analysis, cf. chapter XXII in [Whittaker and Watson, 2013], e.g. we get the PMF

$$(\mathbb{P}(X=k))_{k\in\mathbb{Z}} = \left(\frac{\operatorname{sech}(\pi k)}{\sqrt{2}G}\right)_{k\in\mathbb{Z}}$$

for the first bilateral DHS distribution with Gauß' constant $G = \varpi/\pi$, the lemniscate modulus $k = 1/\sqrt{2}$, and the lemniscate constant $\varpi = \Gamma^2(1/4)/(2\sqrt{2\pi})$.

Characteristic functions, moments, cumulants and some estimators for all six DHS distributions: Due to symmetry, all odd moments of the bilateral DHS distributions vanish. The even moments can be derived using Ramanujan's formulas up to high orders, e. g.

$$\mathbb{E}\left[X^2\right] = \mathrm{Var}(X) = \frac{G^2}{4}.$$

We will calculate every moment generating function and characteristic function which automatically yields a general formula for moments of arbitrary order. We can also determine estimators, e. g. via Pearson's method of moments or maximum likelihood estimation.

New explicit values of series associated with hyperbolic functions: Since the hyperbolic secant is a fix-point of the $-2\pi i$-Fourier transform, viz.

$$\mathcal{F}[x \mapsto \mathrm{sech}(\pi x)](\xi) = \int_{-\infty}^{\infty} \mathrm{sech}(\pi x) e^{-2\pi i x \cdot \xi} \, \mathrm{d}x = \mathrm{sech}(\pi \xi),$$

it is easy to apply the Poisson summation formula to our DHS distributions. In case of the first bilateral DHS distribution, this has enabled us to determine exact values for series of multiplicities of the hyperbolic secant, e. g.

$$\sum_{k \in \mathbb{Z}} \mathrm{sech}^3(\pi k) = \frac{G^3 + G}{\sqrt{2}}.$$

The other distributions will yield more explicit values of hyperbolic series. Further, some of Ramanujan's series identities can be statistically interpreted by our three DHS distributions. E. g. if you randomly generate odd integers with even multiplicity by using the second bilateral DHS distribution, you can calculate the expected outcome up to high order, viz.

$$\mathbb{E}\left[(2Y+1)^8\right] = 7056 G^8.$$

New methods for the integration of special functions in analytic number theory: Using the first bilateral DHS distribution and its associated Fourier series, we can explicitly evaluate a contour integral of a product Ξ of Dirichlet's beta function, the gamma function, and the Riemann zeta function along the critical line $\Re(s) = 1/2$ from Riemann's Hypothesis by using results from [Bourgain, 2016]. We can establish a formula which combines a path contour integral, the lemniscate constant, the lemniscate modulus, Gauß' constant and the variance of the first bilateral DHS, viz. we apply the first hyperbolic secant distribution to analytic number theory. We calculate a contour integral for a product Ξ of Dirichlet's beta function, the gamma function and Riemann's zeta function on the critical line via residues, viz.

$$\int_{1/2-i\cdot\infty}^{1/2+i\cdot\infty} \pi^{-(s+2)}\beta(s+2)\Gamma(s+2)\zeta(s)\,\mathrm{d}s = i\left(\frac{\varpi}{\sqrt{2}}\mathbb{E}\left[X^2\right]-\frac{\pi}{8}\right) = i\left(\frac{\varpi}{\sqrt{2}}\frac{G^2}{4}-\frac{\pi}{8}\right).$$

In this project, we can calculate further exact values of such contour integrals and deepen our understanding of the connections between statistics, special functions, and analytic number theory.

Further generalizations and extensions of the initial DHS distributions: Following the papers [Talacko, 1956] and [Harkness and Harkness, 1968], we can either find a super class of discrete Perks'-type distributions or we can generalise our DHS distributions by adding parameters to the characteristic functions and aim to transfer their continuous techniques to the discrete setting.

3. Statement of the Expected Novelty and Scientific Significance

The introduction of six new discrete hyperbolic secant (DHS) distributions, which have not previously been described in the literature, demonstrates an outstanding degree of novelty in this research. Furthermore, new series values are computed in elementary closed forms. In addition, we are able to statistically interpret parts of Ramanujan's series identities. Finally, a novel connection between statistics and analytic number theory is explored.

The hyperbolic cosine is ubiquitous in nature and constitutes a fundamental function: the average of e^{-x} and e^{x} . Its reciprocal, the hyperbolic secant, therefore finds abundant applications in both theory and practice. Moreover, the fine-structure formulas shed light on the relationship between lemniscate and hyperbolic geometry. On top of that, we are able to elegantly perform complex integration of special functions along the critical line $\Re(s) = 1/2$ by moments of our DHS distributions.

4. Methodology

The project will employ a strictly theoretical and analytical methodology, combining discrete probability theory, Fourier analysis, special functions, and analytic number theory. Its central objective is to define and investigate six new discrete probability distributions based on the hyperbolic secant function, the DHS distributions.

First, the probability mass functions and their normalization constants will be derived from Ramanujan's series identities [Ramanujan and Berndt, 1991]. Their validity will be established by proving positivity, identifying the relevant parameter restrictions, and verifying that the total probability mass equals one. Particular attention will be paid to the distinction between bilateral DHS distributions on $\mathbb{Z}$ and unilateral DHS distributions on $\mathbb{N}_0$, including their support, symmetry, and tail behaviour.

The principal statistical properties of the distributions will then be investigated. These include probability generating functions, characteristic functions, moments, cumulants, variances, and, where appropriate, estimators for the associated parameters. The symmetry of the bilateral distributions will be used to simplify the derivation of their moments and related quantities, with particular emphasis on their even moments.

Fourier-analytic methods will constitute a central part of the investigation. Since the hyperbolic secant function is a fix-point of the $-2\pi i$-Fourier transform, its transform can be used effectively in the analysis of the associated series. The Poisson summation formula will be applied to transform sums over integer lattices into equivalent series with complementary parameters. This will yield explicit evaluations of hyperbolic series and establish connections between the proposed distributions and Ramanujan's identities.

The subsequent analytical work will employ Mellin transformations, Fourier series, and the residue theorem. In particular, contour integrals involving the Dirichlet beta function, the gamma function, and the Riemann zeta function will be studied. These methods will be used to explicitly calculate contour integrals over the important critical line $\Re(s) = 1/2$ from Riemann's Hypothesis. We will explore connections between the discrete hyperbolic secant distributions, special functions, and analytic number theory.

The analysis will combine direct summation, generating-function techniques, Fourier methods, Poisson summation, Mellin-transform methods, residue calculus, asymptotic estimates, and the theory of special functions, cf. [Apostol, 1976] and [Whittaker and Watson, 2013]. Where a closed-form expression cannot be obtained immediately, numerical evidence and special cases will be used to formulate precise conjectures for subsequent proof.

The principal mathematical research material will consist of the six proposed probability mass functions, Ramanujan's series and identities, Fourier series and transforms of hyperbolic functions, the Poisson summation formula, relevant special functions, contour integrals, and existing literature on continuous hyperbolic secant and Perks'-type distributions. No empirical data, human participants, or experimental observations are involved.

Symbolic and numerical computations will be used only as supporting tools. They will assist in identifying conjectures, simplifying expressions, exploring special cases, comparing alternative representations, and verifying numerical approximations. Computational results will be documented together with the relevant assumptions, parameter values, precision, and algorithms. Whenever possible, analytical results will be checked through independent derivations. All central claims and final conclusions will be based on mathematically rigorous proofs.

5. Work Plan and Research Schedule

The project is planned for a period of 48 months and is divided into four overlapping phases. The division provides a clear structure while preserving the flexibility required by exploratory mathematical research.

5.1. Phase I: Foundations and Initial Analysis

Months 1–9

The first phase will establish the mathematical foundations of the project. It will include:

- A detailed review of the literature on continuous hyperbolic secant distributions and their applications, Perks' distributions, and related hyperbolic series.
- The precise definition of the six proposed discrete distributions plus further distributions.
- The derivation and independent verification of their normalisation constants.
- The analysis of their support, symmetry, tail behaviour.
- The formulation of initial conjectures concerning moments and Fourier-analytic identities.

The principal outcome of this phase will be a complete and consistent framework for the subsequent investigation.

5.2. Phase II: Development of the Main Theoretical Results

Months 10–24

The second phase will focus on the development and proof of the principal mathematical results. The work will include:

- The derivation of probability generating functions and characteristic functions.
- The calculation of moments, cumulants, and variances.

- The construction and analysis of suitable estimators.
- The formulation of central conjectures and intermediate propositions.
- The derivation of auxiliary lemmas and structural results.
- The application of the Poisson summation formula to hyperbolic series.
- The comparison of the resulting identities with Ramanujan's formulas.

The objective of this phase is to establish the core theoretical results and to determine the precise conditions under which they are valid.

5.3. Phase III: Extensions, Generalisations, and Validation

Months 25–39

The third phase will examine the robustness and generality of the results obtained in Phase II. In particular, it will address:

- Extensions to broader families of discrete hyperbolic distributions.
- Generalisations involving additional scale or location parameters.
- The study of boundary cases and exceptional configurations.
- Further explicit evaluations of hyperbolic series.
- The investigation of connections with Dirichlet's beta function, the gamma function, and Riemann's zeta function.
- Comparisons with existing probability distributions and known statistical models.
- High-precision numerical verification of the analytical results.

Any computational work will serve as a complementary tool for exploring examples, testing conjectures, and identifying relevant patterns. The mathematical validity of the results will remain based on rigorous analytical proofs.

5.4. Phase IV: Consolidation, Dissemination, and Completion

Months 40–48

The final phase will consolidate the results and prepare them for dissemination. It will include:

- The completion and verification of all proofs.
- The integration of the individual results into a coherent theoretical framework.
- The preparation and submission of research articles.
- The presentation of results at conferences, workshops, and seminars.
- The preparation of supplementary symbolic and numerical materials.
- The preparation of the final project report.

Particular attention will be paid to ensuring that the final results are clearly positioned within the international research literature. Furthermore, the results will be disseminated through several complementary channels:

- Publication in peer-reviewed international journals.
- Presentations at international conferences and specialised workshops.
- Invited seminars at universities and research institutions.
- Formal and informal exchanges with researchers working in related fields.
- Deposition of preprints and, where appropriate, mathematical software or code in recognised repositories.

The project is expected to result in approximately three to five research articles. Manuscripts will be prepared continuously throughout the project rather than only at its conclusion. Preliminary results will be shared with the research community once they have reached an appropriate level of mathematical maturity.

6. Project-relevant Scientific Qualifications of the Researchers Involved

TBA [...]

Dr. Leonard Pleschberger has completed comprehensive curricula in probability (time-continuous stochastic processes) and analysis (PDEs and Harmonic Analysis). He obtained his doctorate on the fractal geometry of stochastic processes under the supervision of Prof. Dr. Peter Kern at Heinrich Heine University Düsseldorf in Germany. Together with Prof. Kern, he published the article "Parabolic Fractal Geometry of Stable Lévy Processes with Drift" in the *Journal of Fractal Geometry*, cf. [Kern and Pleschberger, 2024]. He also worked in the field of mathematical semi-conductor physics: He prove Lifshitz tails for fractional random Schrödinger operators with Gaußian potential by using stable Lévy processes in "On the Integtrated Density of States of Fractional Random Schrödinger Operators", cf. [Kern and Pleschberger, 2025]. Further he developed some "Dirichlet-Neumann Bracketing for Fractional Random Schrödinger Operators", cf. [Pleschberger, 2025]. In addition, he holds Bachelor's degrees in Philosophy & Ancient Culture, and a Bachelor's degree in Fashion- & Design Management. During his doctorate, he gained experience leading teams of up to 10 members and designed four volumes of proceedings for an oncology conference. He also held various talks about the parabolic fractal geometry of stochastic processes and its connections to the Navier-Stokes equations in Düsseldorf, Cologne, Bielefeld, Darmstadt, Klagenfurth, Berlin, Vienna and on Proquerolles in South-France.

7. Ethical, Safety-Related and Regulatory Aspects

The proposed project is theoretical mathematical research. It does not involve human participants, animals, clinical interventions, personal or sensitive data, biological materials, or physical experiments. Consequently, no substantial ethical, health and safety, or regulatory issues are expected to arise.

The project does not involve classified information, critical infrastructure, dual-use technologies, or the development of software intended for harmful applications. Any computational work will use standard mathematical software and publicly available or institutionally licensed resources. No personal data will be collected, processed, or stored.

The research will nevertheless be conducted in accordance with the applicable institutional and legal requirements. Particular attention will be paid to research integrity, accurate attribution of previous results, avoidance of plagiarism, transparent documentation of methods, and reproducibility of computational calculations where applicable. Software, datasets, and other third-party materials will be used only in accordance with their respective licence conditions.

If the scope or methodology of the project changes in a way that introduces ethical, security-related, data-protection, or regulatory concerns, the applicant will consult the responsible institutional bodies and obtain any required approvals before proceeding.

8. Sex and Gender Dimensions of the Project

While the core research questions of this project are methodological and theoretical in nature, gender-relevant considerations are addressed in the project's team composition, supervision practices, and communication strategy, ensuring an inclusive and equitable research environment. The project team is committed to gender-equitable and diversity-conscious practices throughout the project's life-cycle. This includes the following measures:

Team composition and recruitment: Should additional project staff (e.g., research assistants, doctoral or master's students) be recruited during the course of the project, recruitment procedures will be conducted according to principles of equal opportunity, with job advertisements formulated in a gender-neutral and inclusive manner to encourage applications from underrepresented groups in mathematics and the natural sciences.

Supervision and mentoring: Supervision practices within the project will be designed to provide equal support and visibility to all team members, irrespective of gender, and to foster an inclusive working environment.

Dissemination and communication: In presenting project results at conferences, in publications, and in outreach activities, the project team will be mindful of balanced representation (e.g., in the choice of collaborators, invited speakers, or acknowledged contributors) wherever feasible within the scope of the project.

Although the mathematical subject matter of this project does not itself involve human subjects, data, or applications with direct sex- or gender-specific implications, the applicant considers it important to reflect on gender and diversity dimensions in the broader research process and academic culture surrounding the project.

A. Appendix 1: References

[Apostol, 1976] Apostol, Tom M. (1976). *Introduction to Analytic Number Theory*. Springer. DOI: 10.1007/978-1-4757-5579-4

[Bentley, 1975]Bentley, Jon L. (1975). *Multidimensional binary search trees used for associative searching*. Communications of the ACM, Volume 18(9), pp. 509–517. DOI: 10.1145/361002.361007

[Bourgain, 2016] Bourgain, Jean L. (2016). *Decoupling, exponential sums and the Riemann zeta function*. DOI: 10.48550/arXiv.1408.5794

[De Bruijn, 1958]De Bruijn, N. G. (1958). *Asymptotic Methods in Analysis*. Dover Publications. ISBN 978-0-486-64221-5

[Fischer, 2014] Fischer, Matthias J. (2014). *Generalized Hyperbolic Secant Distributions With Applications to Finance*. Springer. DOI: 10.1007/978-3-642-45138-6

[Fisher, 1921] Fisher, Ronald A. (1921). *On the "Probable Error" of a coefficient of correlation deduced from a small sample*. Metron 4, I, pp. 3–32.

[Flajolet and Puech, 1986] Flajolet, Philippe; Puech, Claude (1968). *Partial match retrieval of multidimensional data*. Journal of the ACM (JACM), Volume 33(2), pp. 371–407. DOI: 10.1145/5383.5453

[Flajolet, Gourdon and Dumas, 1995] Flajolet, .P; Gourdon, X.; Dumas, P. (1995). *Mellin Transforms and Asymptotics: Harmonic Sums*. Theoretical Computer Science, 144, pp. 3–58. DOI: 10.1016/0304-3975(95)00002-E

[Harkness and Harkness, 1968] Harkness, W. L.; Harkness, M. L. (1968). *Generalized Hyperbolic Secant Distributions*. Journal of the American Statistical Association, 63(321), pp. 329–337. DOI: 10.1080/01621459.1968.11009246

[Jiu and Peng, 2025] Jiu, Lin; Peng, Linyu (2025). *Information geometry and α-parallel prior of the beta-logistic distribution*. Communications in Statistics: Theory and Methods 54, 3292–3306. DOI: 10.1080/03610926.2024.2387839

[Kern and Pleschberger, 2024] Kern, Peter; Pleschberger, Leonard J. (2024). *Parabolic Fractal Geometry of Stable Lévy Processes with Drift*. J. Fractal Geom. 11, no. 3/4, pp. 343–371. DOI: 10.48550/arXiv.2312.13800

[Kern and Pleschberger, 2025] Kern, Peter; Pleschberger, Leonard J. (2025). *On the Integrated Density of States of Fractional Random Schrödinger Operators*. Via GitHub.

[Kirschenhofer, Prodinger and Szpankowski, 1996] Kirschenhofer, P.; Prodinger, H.; Szpankowski, W. (1996). *Multidimensional Digital Searching and Some New Parameters in Tries*. International J. Foundation of of Computer Science, 4, 69–84.

[Morris, 1982] Morris, Carl N. (1982). *Natural Exponential Families with Quadratic Variance Functions*. Ann. Statist. 10(1), pp. 65–80. DOI: 10.1214/aos/1176345690

[Morris, 1983] Morris, Carl N. (1983). *Natural Exponential Families with Quadratic Variance Functions: Statistical Theory*. Ann. Statist. 11(2), pp. 515–529. DOI: 10.1214/aos/1176346158

[Pleschberger, 2025] Pleschberger, Leonard J. (2025). *Dirichlet-Neumann Bracketing for Fractional Random Schrödinger Operators*. Via GitHub.

[Ramanujan and Berndt, 1991] Ramanujan, Srinivasa; Berndt, Bruce, ed. (1991). *Ramanujan's Notebooks. Part III*. Springer. DOI: 10.1007/978-1-4612-0965-2

[Rivest, 1976] Rivest, Ronald L. (1976). *Partial-Match Retrieval Algorithms*. SIAM Journal on Computing, 5(1). DOI: 10.1137/0205003

[Szpankowski, 2001] Szpankowski, Wojciech (2001). *Average Case Analysis of Algorithms on Sequences*. John Wiley & Sons. DOI: 10.1002/9781118032770

[Talacko, 1956] Talacko, J. (1956). *Perks' Distributions and Their Role in the Theory of Wiener's Stochastic Variables*. Trabajos de Estadística, 7, pp. 159–174. DOI: 10.1007/BF03003994

[Whittaker and Watson, 2013] Whittaker, Edmund T.; Watson, George N. (2013). *A Course of Modern Analysis. An introduction to the general theory of infinite processes and of analytic functions with* *an account of the principal transcendental functions (4th edition)*. Cambridge University Press. DOI: 10.1017/CBO9780511608759

B. Appendix 2: Host Research Institution and Justification of Requested Funds

The project will be carried out at the Department of Statistics and Operations Research at the University of Vienna, which will serve as the host research institution. The institution provides the necessary academic environment and infrastructure for the proposed theoretical research, including access to mathematical literature, electronic journals, library resources, office facilities, and standard computational equipment.

No associated research institution is involved in the project. The research will be conducted entirely within the host institution, and no external laboratory, experimental facility, or specialised infrastructure is required.

The requested funds are necessary to ensure the effective implementation of the project. Personnel costs constitute the main budget item, as the project requires sustained research activities over the proposed project period. Additional funds are requested, where applicable, for participation in conferences and research visits, which will support the presentation of results, scientific exchange, and collaboration with researchers working in related areas. Costs for mathematical software, computing resources, and publication may also be required to support symbolic calculations, numerical verification, and the dissemination of the research findings.

All requested funds are directly related to the objectives and work programme of the project and will be used economically and in accordance with the applicable institutional and funding regulations.

C. Appendix 3: Scientific CVs and Overview of Previous Research Achievements

D. Appendix 4: Statement on Environmental Sustainability in the Research Process

The project will be conducted in accordance with the principles of environmental sustainability and resource efficiency. As the research is primarily theoretical and computational, it does not require laboratory experiments, fieldwork, or the use of hazardous substances. Consequently, its direct environmental impact is expected to be limited. Nevertheless, the project team will take the following measures to reduce its environmental footprint:

Computational analyses will be planned efficiently, and unnecessary simulations and repeated calculations will be avoided; data, code, and intermediate results will be stored and managed in a structured manner to reduce redundant storage and data transfers; existing software, datasets, and institutional infrastructure will be used whenever possible; online meetings and digital collaboration will be preferred where appropriate; necessary travel will be limited and, whenever feasible, public transport will be used; purchases will be restricted to essential equipment and services, with consideration given to durability, energy efficiency, and responsible procurement; digital dissemination will be prioritized, while printed materials will be kept to a minimum.

Environmental sustainability will therefore be considered throughout the entire research process, including project planning, data collection and analysis, collaboration, dissemination, and resource management. Where appropriate, the project's practices will be reviewed during its implementation and adjusted to further reduce resource consumption.